

NETWORK SIMULATION LAB

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS17	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OVERVIEW:

The course is designed for the students to develop their network simulation skills using NS3. It will help the students to work with their networking and research projects.

COURSE OBJECTIVES:

1. Understand the importance and usage of NS3
2. To inculcate working knowledge among Node to Node communication in networks.
3. Understand and implement various networking topology.
4. To familiarize with the Net Anim

COURSE OUTCOMES:

By the end of this course, the student will be able to

1. Install NS3 and its supporting packages.
2. Implement connecting various nodes
3. Implement Star and bus and Hybrid topologies using NS3
4. Use and develop applications using NetAnim.

SYLLABUS

EXPERIMENTS:

1. Introduction about discrete events simulation and its tools.
2. Installation of NS3 in Linux Operating System.
3. Program in NS3 to connect two nodes.
4. Program in NS3 for connecting three nodes considering one node as a central node.
5. Implement physical star topology and simulate using NS3.
6. Implement physical bus topology and simulate using NS3.
7. Program for connecting multiple routers and nodes and building a hybrid topology.
8. Installation and configuration of NetAnim
9. Program in NS3 to implement FTP using TCP bulk transfer.
10. Implement AODV Routing Protocol
11. Implement DSDV Routing Protocol.
12. Implement DSR Routing Protocol.

TEXT BOOKS:

1. Andrew S Tanenbaum David J Wetherall , Computer Networks, 5th Edition, Pearson Education.
2. Kurose, Ross (2010), Computer Networking: A top down approach, Sixth Edition, Pearson Education, India.

REFERENCES:

1. Behrouz A. Forouzan, Data communication and Networking, 4th Edition, Mc Graw-Hill, India.
2. Larry L. Peterson, Bruce S. Davie, Computer Networks: A Systems Approach, MK Publishers

WEB LINKS:

1. <https://www.nsnam.org/releases/>
2. <https://www.nsnam.org/documentation/>